

Using Disjunctions to Warm Start MILPs

Ph.D. Dissertation Defense

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Outline

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- 2 Notation and Preliminaries
- 3 Paper 1: Parametric Disjunctive Cuts
- 4 Paper 2: Strengthening PDIs
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Problem Framing and Big Idea

Sequential MILPs appear throughout optimization

- decomposition methods
 - branch and price
 - bilevel optimization
- online optimization
 - unit commitment

Solving each instance from scratch wastes reusable structure [2–5, 7]

- partial solutions
- pseudocosts
- cut generation and selection rules
- branching decisions (disjunctions)

Warm starting

Reuse structural information from previous solves to solve a related MILP more quickly.

Big idea

Disjunction generation is central to solving MILPs, but is largely overlooked in warm starting. This dissertation studies how to reuse that structure.

Defining Disjunctions and Their Warm Start

Disjunction

A disjunction is a union of logical statements; in branch and bound, it manifests as the branching decisions that induce the leaves of a partial branch-and-bound tree.

To warm start, apply the branching decisions from a previously explored partial tree to a related MILP instance to induce a corresponding partial tree. Then pick a reuse mode:

Explicit reuse (a.k.a. node warm start)

Initialize the node pool of the new solve with the leaves of the induced partial tree.

Implicit reuse (a.k.a. cut warm start)

Initialize the cut pool with *disjunctive inequalities* valid for the union of the leaves induced by reused branching decisions.

Core contrast

Node warm starts transfer the tree directly; cut warm starts transfer cuts that capture its effect.

Why This Is Hard, and Why It Is Timely

- **Explicit reuse can be brittle.** Large inherited trees can cause significant excess searching, particularly in longer solves.
- **Generating cuts against large disjunctions was intractable.** VPC-based disjunctive cut generation significantly reduces the size of the cut generating LP.
- **Solving LPs to generate cuts is still not ideal.** The strength of disjunctive cuts often does not offset their cost of generation.

Opportunity

If disjunctive cut generation can be made cheap enough, implicit reuse may be a viable way to retain useful branching information without inheriting a large node pool and its excess searching.

Thesis

Disjunctions can effectively warm start both branching and cutting in branch and cut.

- ① **Preliminaries:** define disjunctions, bounding functions, and certificates.
- ② **Paper 1:** build parametrically valid disjunctive cuts for cut warm starts.
- ③ **Paper 2:** strengthen those cuts when they weaken across instances.
- ④ **Paper 3:** compare node warm starts and cut warm starts under shared disjunctions.

MILP and Disjunction Basics

MILP family

$$\min_{x \in S^k} c^k x, \quad S^k = \{x \in \mathcal{P}^k : x_j \in \mathbb{Z} \ \forall j \in \mathcal{I}\}, \quad \mathcal{P}^k = \{x \in \mathbb{R}^n : A^k x \geq b^k\}.$$

- **Definition 1 (valid inequality).** (α, β) is valid for $S \subseteq \mathbb{R}^n$ if $S \subseteq \{x \in \mathbb{R}^n : \alpha^\top x \geq \beta\}$.
- **Definition 2 (valid disjunction).** $\mathcal{X} = \{\mathcal{X}^t\}_{t \in T}$ is valid for S if $S \subseteq \cup_{t \in T} \mathcal{X}^t$.

Disjunctive terms

$$\mathcal{X}^t := \{x \in \mathbb{R}^n : D^t x \geq D_0^t\}, \quad Q^{kt} = \mathcal{P}^k \cap \mathcal{X}^t = \{x \in \mathbb{R}^n : A^{kt} x \geq b^{kt}\}.$$

For our purposes, the constraints in $\mathcal{X}^t$ represent the branching constraints defining term t .

- **Definition 3 (disjunctive hull).** $\mathcal{P}_D^k = \text{cl conv}(\cup_{t \in T} Q^{kt})$.
- **Definition 4 (valid disjunctive inequality).** (α, β) is a valid disjunctive inequality if it is valid for $\mathcal{P}_D^k$.
- **Definition 5 (feasible basis).** A basis is any index set of n linearly independent constraints binding at a basic solution; it is feasible if that basic solution lies within the polyhedron Q^{kt} .

Branch and bound

- Solve LP relaxations over subproblems created by branching on fractional variables.
- Integer-feasible solutions improve the best known objective value and help prune subproblems.
- LP relaxations bound the best objective value each subproblem can still achieve.

Cutting plane method

- Strengthen an LP relaxation by generating valid inequalities for the integer-feasible set.
- This tightens a node relaxation without branching.

Branch and cut

- Modern MILP solvers combine both: cuts tighten node relaxations, and branching resolves the remaining integrality.

MILP Duality

Branch and cut evaluates the optimal-value function at one instance in a sequence of related MILPs

Value function

$$\phi(k) = \inf\{c^k x : x \in \mathcal{S}^k\}$$

by progressively tightening dual (lower) and primal (upper) bounding functions until they meet.

Dual bounding function

$$F_\theta : \mathcal{K} \rightarrow \mathbb{R} \cup \{-\infty\} \text{ with } F_\theta(k) \leq \phi(k)$$

Primal bounding function

$$G_\theta : \mathcal{K} \rightarrow \mathbb{R} \cup \{+\infty\} \text{ with } \phi(k) \leq G_\theta(k)$$

Warm Starting

Key insight

The primal and dual bounding functions are defined over the entire MILP series, so plugging in the data of a new instance immediately gives valid bounds.

Concretely, this includes the following forms.

- **Dual bound from a disjunction:**

$$F_{\{\mathcal{X}^t\}_{t \in T}}(k) = \min_{t \in T} \min_{x \in Q^{kt}} c^k x.$$

- **Dual bound from a cut pool $\Pi = \{(\alpha^w, \beta^w)\}_{w \in W}$:**

$$F_{\Pi}(k) = \min\{c^k x : x \in \mathcal{P}^k, \alpha^{w\top} x \geq \beta^w \ \forall w \in W\}.$$

- **Primal bound from a candidate x :** $G_x(k) = c^k x$ if $x \in \mathcal{S}^k$, else $+\infty$.

MILP duality connection to (dual) warm starting

The first dual bound for a new instance is obtained by re-evaluating a previously constructed disjunction- or cut-based dual bounding function with the new data; all later dual bounds in that solve refine this starting point.

Certificates of Validity for Disjunctive Cuts

- **Definition 6 (certificate of validity).** $\{v^t\}_{t \in T}$ certifies (α, β) for MILP k if $\alpha_j \geq \max_t \{v^t A_{.j}^{kt}\}$ for $j \in \mathcal{I}$ and $\beta \leq \min_t \{v^t b^{kt}\}$.
- A **Farkas certificate** is a certificate of validity that satisfies Farkas' lemma (FL).
- Kazachkov and Balas [6] show how to recover Farkas certificates for {cut, disjunction, MILP instance} triples.

$$\left. \begin{array}{l} \alpha^\top = v^t A^{kt} \\ \beta \leq v^t b^{kt} \\ v^t \in \mathbb{R}_+^{1 \times (q+q_t)} \end{array} \right\} \quad \text{for all } t \in T \quad (\text{FL})$$

Relating Certificates to Their Generators

- **Definition 7 (induce).** IP- k and $\mathcal{X}$ induce $\{v^t\}_{t \in T}$ if $v^{t_1} A^{kt_1} = v^{t_2} A^{kt_2}$ for all feasible term pairs.
- **Definition 8 (determine).** Basis $\mathcal{A}^t$ determines v^t if $\{i : v_i^t > 0\} \subseteq \mathcal{A}^t$.

Why we care

Certificates encode why a disjunctive cut is valid, so they are the right objects to reuse implicitly across instances. We build the mechanism for doing this in Paper 1.

Paper 1: PDIs

- **Definition 9 (PDI).** Let $\mathcal{K}$ index a family of MILPs, let $\{\mathcal{X}^t\}_{t \in T}$ be valid for every member of the family, and let $\{v^t\}_{t \in T}$ be nonnegative. Then $\{(\alpha^k, \beta^k)\}_{k \in \mathcal{K}}$ is a family of PDIs if, for every $k \in \mathcal{K}$, $\{v^t\}_{t \in T}$ is a certificate of validity for (α^k, β^k) relative to $\mathcal{P}_D^k$.
- **Theorem 1.** Let $k, \ell \in \mathcal{K}$. Given a Farkas certificate $\{v^t\}_{t \in T}$ induced by disjunction $\{\mathcal{X}^t\}_{t \in T}$ and IP- k , we have $\alpha^\top x \geq \beta$ for all $x \in \mathcal{P}_D^\ell$, where

$$\alpha_j = \max_t \{v^t A_{.j}^{\ell t}\}, \quad \beta = \min_t \{v^t b^{\ell t}\}$$

for all $j \in [n]$.

Math to English

Reuse the old certificate, plug in the new instance data, and you still get a cut valid for the new disjunctive hull.

Why we care

This gives a principled way to *warm start the cut pool* by reusing certificate information instead of rebuilding the cut from scratch.

Paper 1: A Visual Example

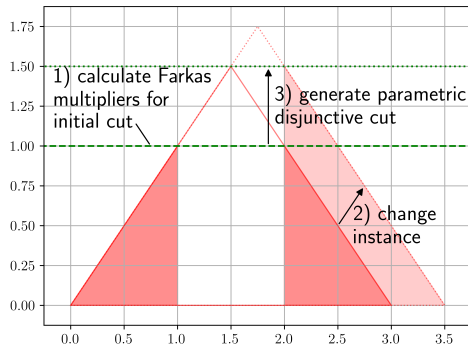


Figure Explanation

Apply Theorem 1 to 1) the certificate from the dashed cut and 2) a perturbed MILP instance to yield 3) the dotted valid inequality for the perturbed instance.

Paper 1: Supporting PDIs

- Validity alone is not enough; a weak valid cut may not matter computationally.
- **Definition 11 (support).** (α, β) valid for polyhedron $\mathcal{P}$ supports $\mathcal{P}$ if there exists $x \in \mathcal{P}$ with $\alpha^\top x = \beta$.
- **Lemma 1.** Let $k, \ell \in \mathcal{K}$; let $\{v^t\}_{t \in T}$ be induced by IP- k and $\{\mathcal{X}^t\}_{t \in T}$; let $\{(\alpha^k, \beta^k)\}_{k \in \mathcal{K}}$ be the associated family of Farkas PDIs; and let $\mathcal{A}^t$ determine v^t for all t . Then $(\alpha^\ell, \beta^\ell)$ supports $\mathcal{P}_D^\ell$ if
 - ① $\{v^t\}_{t \in T}$ is induced by IP- ℓ and $\{\mathcal{X}^t\}_{t \in T}$, and
 - ② $\mathcal{A}^t$ is feasible for $Q^{\ell t}$ for all $t \in T$.

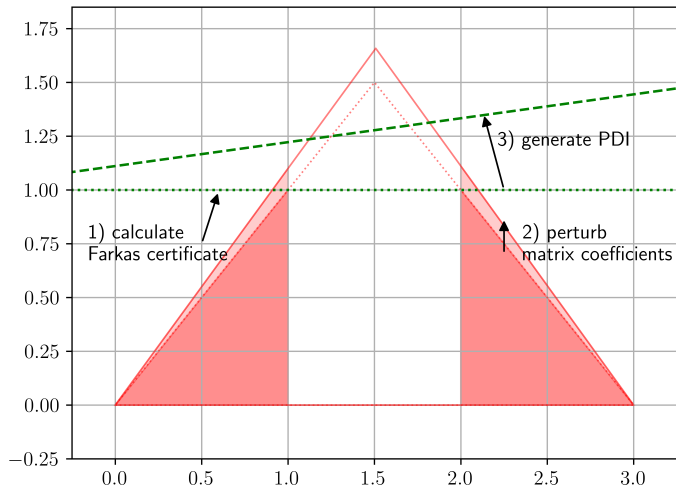
Math to English

If perturbation does not change the coefficient matrix or basis feasibility status, the regenerated cut maintains disjunctive hull support.

Why we care

If PDI performance is poor, this lets us distinguish whether the idea failed or whether the reused certificate simply lost support under perturbation.

Paper 2: Coefficient Matrix Changes



Dissertation Figure 4.1

- PDIs can stay valid after perturbation
- but lose support for the disjunctive hull

Interpretation

This is the failure mode Paper 2 addresses.

Paper 2: PDIs May Lose Disjunctive Hull Support

- **Lemma 2.** Let $\ell \in \mathcal{K}$, $t \in \mathcal{T}$, and $\alpha \in \mathbb{R}^n$. If $\mathcal{Q}^{\ell t}$ is bounded and nonempty, then $(\alpha, \alpha^\top \bar{x})$ with $\bar{x} \in \arg \min_{x \in \mathcal{Q}^{\ell t}} \alpha^\top x$ is a supporting inequality for $\mathcal{Q}^{\ell t}$ with supporting certificate

$$\bar{v} \in \arg \max_{v \in \mathcal{D}^{\ell t}(\alpha)} v b^{\ell t}, \quad \mathcal{D}^{\ell t}(\alpha) = \{v : v A^{\ell t} = \alpha, v \geq 0\}.$$

- **Corollary 1.** If no row of $A^{\ell t}$ is parallel to α , then that supporting point/certificate pair is unique.

Math to English

For fixed cut coefficients, a LP can be solved to determine a bound such that it supports the disjunctive hull.

Why we care

If a reused PDI lacks disjunctive hull support, Lemma 2 gives a repair mechanism that restores the strongest cut with the same direction. Corollary 1 shows solving an LP is often necessary.

Paper 2: Strengthening Algorithm

Algorithm 1 Disjunctive Hull Supporting Cut Generator

Require: $k, \ell, \{\mathcal{X}^t\}_{t \in T}, \{v^t\}_{t \in T}, \{\mathcal{A}^t\}_{t \in T}$

Ensure: IP- k and $\{\mathcal{X}^t\}_{t \in T}$ induce $\{v^t\}_{t \in T}$

Ensure: $\{\mathcal{A}^t\}_{t \in T}$ determines $\{v^t\}_{t \in T}$

Ensure: There exists $t \in T$ such that $\mathcal{A}^t$ is feasible for $Q^{\ell t}$ and $v^t \neq 0$

- 1 $T_f \leftarrow \{t \in T : v^t \neq 0, (A_{\mathcal{A}^t}^{\ell t})^{-1} b_{\mathcal{A}^t}^{\ell t} \in Q^{\ell t}\}$
- 2 $\{\bar{v}^t\}_{t \in T} \leftarrow \{v^t\}_{t \in T}$
- 3 $(\alpha, \beta) \leftarrow \text{Theorem1}(\ell, \{\mathcal{X}^t\}_{t \in T_f}, \{v^t\}_{t \in T_f})$
- 4 For all $t \in T_{\text{feas}}^{\ell}$ such that $A^{kt} \neq A^{\ell t}$ or $t \notin T_f$, set

$$\bar{v}^t \leftarrow \arg \max_{y \in \mathcal{D}^{\ell t}(\alpha)} \{y^{\top} b^{\ell t}\}$$

- 5 $(\alpha, \bar{\beta}) \leftarrow \text{Theorem1}(\ell, \{\mathcal{X}^t\}_{t \in T_{\text{feas}}^{\ell}}, \{v^t\}_{t \in T_{\text{feas}}^{\ell}})$
 - 6 **return** $(\alpha, \bar{\beta})$
-

Paper 2: Strengthening Algorithm Supports Disjunctive Hull

Main guarantee

Theorem 2. Let $k, \ell \in \mathcal{K}$, let $\{\mathcal{X}^t\}_{t \in T}$ be a disjunction, let $\{v^t\}_{t \in T}$ be Farkas multipliers induced by IP- k and $\{\mathcal{X}^t\}_{t \in T}$, and let $\mathcal{A}$ determine $\{v^t\}_{t \in T}$. If there exists $t \in T$ such that $\mathcal{A}^t$ is feasible for $Q^{\ell t}$ and $v^t > 0$, then the strengthened generator returns (α, β) supporting

$$\text{cl conv} \left(\bigcup_{t \in T} Q^{\ell t} \right).$$

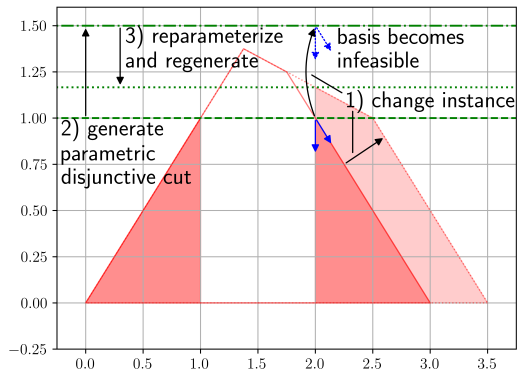
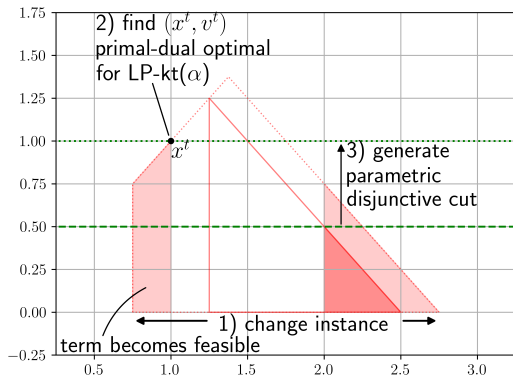
Math to English

Any previously valid certificate can be turned into a hull-supporting cut, provided at least one previously feasible term basis remains feasible after perturbation.

Why this matters

This is the formal guarantee that the repair step is not just heuristic: after reparameterization, the regenerated cut is guaranteed to recover disjunctive hull support.

Paper 2: Strengthening Algorithm Visualized



Takeaway

When feasible bases become infeasible or infeasible bases become feasible, PDI's can lose disjunctive hull support but recover it with Algorithm 1.

Paper 2: Experimental Setup

Test-set generation

From **Base** instances (Presolved MIPLIB 2017; $< 5,000 \times 5,000$), we build **Test** sets indexed by (k, u, θ) , where **Element** perturbed $= u \in \{A, b, c\}$ and **Degree** of perturbation $= \theta$. Experiments vary disjunctive cut generation method and size ($d \in \{4, 64\}$ **Terms**).

Methods compared

We solve each test set under each of the following configurations

- **VPC**: nonparametric $\mathcal{V}$ -polyhedral disjunctive cuts
- **PDI**: parametric disjunctive inequalities
- **S-PDI**: strengthened PDIs
- **Default**: default solver with no disjunctive cuts

Paper 2: Integrality Gap Closed

Root-gap metric from the dissertation appendix

Let x_I denote an optimal MILP solution, let $\bar{x}$ denote an optimal LP-relaxation solution before adding disjunctive information, and let x_0 denote the dual bound after adding cuts or applying a disjunction.

$$\% \text{ integrality gap closed} = 100 \times \frac{c^\top x_0 - c^\top \bar{x}}{c^\top x_I - c^\top \bar{x}}.$$

Math to English

The denominator is the original LP-to-MILP gap. The numerator is how much the dual bound changed.

How to read subsequent tables

Larger values mean the dual bound after applying a disjunction and/or cuts moved closer to the MILP optimum. I.e. a value of 0% means the dual bound did not improve upon the LP relaxation, while 100% would mean the full integrality gap was closed.

Paper 2: S/PDIs Strengthen LP Relaxations

Degree	Terms	Element	VD	PD	VPC	S-PDI	PDI
0.5	4	matrix	5.96	2.67	3.38	1.90	1.90
0.5	4	objective	5.37	4.74	3.60	–	3.18
0.5	4	rhs	8.58	3.69	3.32	2.23	2.23
0.5	64	matrix	12.61	7.88	5.94	3.51	3.31
0.5	64	objective	13.70	12.09	7.88	–	6.21
0.5	64	rhs	17.66	9.99	11.16	6.47	5.21
2.0	4	matrix	8.33	2.53	4.30	1.10	1.10
2.0	4	objective	6.06	4.35	4.21	–	2.83
2.0	4	rhs	18.63	3.00	3.14	0.48	0.48
2.0	64	matrix	16.26	5.75	7.43	1.18	0.44
2.0	64	objective	15.59	10.87	8.52	–	5.22
2.0	64	rhs	27.38	7.31	9.13	1.16	1.08

VD = VPC disjunctive dual bound, PD = S/PDI disjunctive dual bound.

Increased parameterization and perturbation weaken disjunctive cuts, but S-PDI and PDI still improve LP relaxation.
Weak improvements can occur when cuts focus deeper in tree [1].

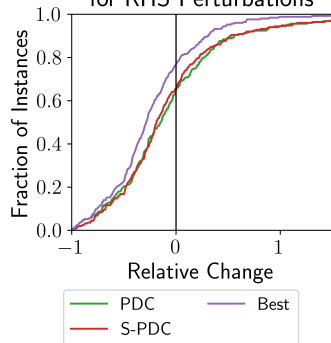
Paper 2: S/PDIs Reduce Root Node Processing Time

Degree	Terms	Element	VPC	S-PDI	PDI	Default
0.5	4	matrix	1.394	0.480	0.435	0.314
0.5	4	objective	1.893	–	0.601	0.572
0.5	4	rhs	0.690	0.297	0.300	0.275
0.5	64	matrix	11.878	2.135	1.061	0.336
0.5	64	objective	19.130	–	1.309	0.530
0.5	64	rhs	6.428	0.744	0.549	0.313
2.0	4	matrix	2.450	0.377	0.373	0.296
2.0	4	objective	1.758	–	0.631	0.533
2.0	4	rhs	0.794	0.248	0.234	0.239
2.0	64	matrix	9.564	1.704	1.211	0.288
2.0	64	objective	19.378	–	1.408	0.540
2.0	64	rhs	4.358	0.546	0.515	0.234

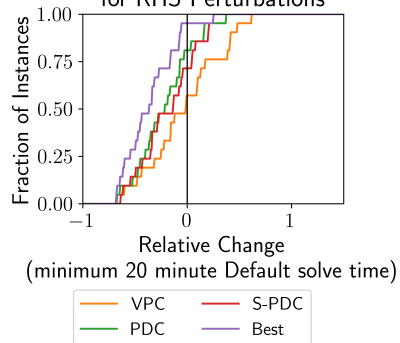
Increased parameterization decreases root node processing time.

Paper 2: Branch and Cut Termination Time for RHS Perturbations

Solve Time Change Relative to VPC
for RHS Perturbations



Solve Time Change Relative to Default
for RHS Perturbations



PDIs often outperform VPCs (left) and almost always beat default Gurobi (right).

Paper 3: Experimental Setup

Test-set generation

Base instances are pulled from MIPLIBs 3, 2003, and 2010. Test set are generated in the same fashion as before.

Methods compared

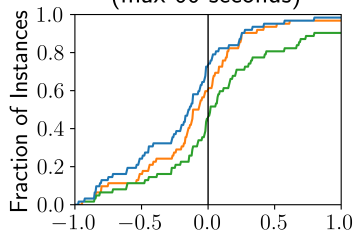
We solve each test set under each of the following configurations

- **NWS**: node (explicit) warm start via reused partial branch-and-bound leaves
- **CWS**: cut (implicit) warm start via parametrically generated disjunctive cuts
- **Default**: default solve with no warm start

We use the same disjunction to initialize both warm-starting methods.

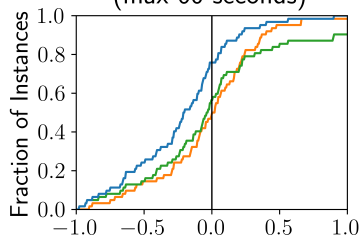
Paper 3: Branch and Cut Performance - Short Solves

Change in Nodes Processed
Relative to Default
(max 60 seconds)



— Cuts Warm Start — Best
— Nodes Warm Start

Change in LP Iterations
Relative to Default
(max 60 seconds)



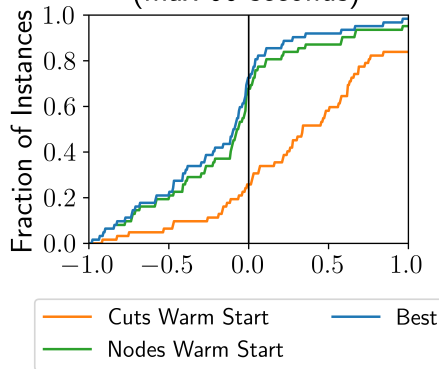
— Cuts Warm Start — Best
— Nodes Warm Start

CWS replaces some branching and shrinks trees, but it also enlarges LP relaxations.

NWS reuses warm-started bases, so it more reliably cuts LP iterations.

Paper 3: Branch and Cut Performance - Short Solves

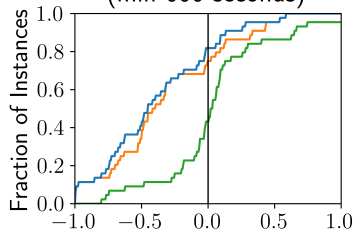
Change in Solve Time
Relative to Default
(max 60 seconds)



- CWS makes LPs denser, and for short solves the tree is not reduced enough to compensate, so solve time can increase.
- NWS keeps LPs at essentially the same size and density but benefits from fewer pivots via warm-started bases.
- Result: NWS is the better short-solve regime.

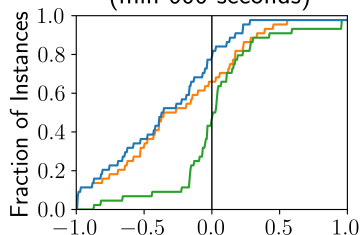
Paper 3: Branch and Cut Performance - Long Solves

Change in Nodes Processed
Relative to Default
(min 600 seconds)



— Cuts Warm Start — Best
— Nodes Warm Start

Change in LP Iterations
Relative to Default
(min 600 seconds)



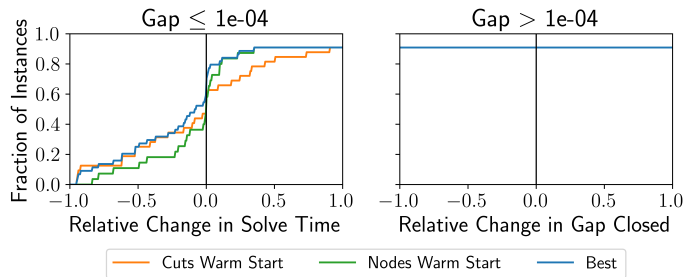
— Cuts Warm Start — Best
— Nodes Warm Start

CWS can prune large subtrees, so much smaller trees can outweigh larger LPs.

NWS warm starts only a small prefix of the eventual terminal tree, so its effect is limited.

Paper 3: Branch and Cut Performance - Long Solves

Change in Solve Performance Relative to Default
(min 600 seconds)



- For longer solves, CWS has more opportunity to prune enough nodes that the harder LPs are worth it.
- Much smaller trees can outweigh larger LP relaxations, so CWS wins on a larger share of instances.
- NWS warm starts a small tree relative to the eventual terminal tree, so it often has little effect.

Conclusion and Outlook

What We Contributed

- A unified view of disjunction reuse in branch and cut.
- How cut certificates yield valid and disjunctive-hull-supporting parametric disjunctive cuts.
- A controlled comparison between cut and node warm starts under shared disjunctions.

What We Learned

Warm-start benefit is fundamentally regime-dependent:

- Cut warm starts tend to create denser, larger LPs but can substantially shrink the tree.
- This better suits node warm starts to shorter solves and cut warm starts to longer solves.

Open Questions

- Can certificate recovery be extended to other expensive cut families?
- Can we reduce the density of disjunctive cuts without losing their pruning power?
- When exactly should a solver prefer node reuse, cut reuse, or a hybrid strategy?

References I

- [1] Egon Balas and Aleksandr M. Kazachkov. $\mathcal{V}$ -polyhedral disjunctive cuts, 2025. URL <https://doi.org/10.1007/s10107-025-02227-y>.
- [2] Arnaud Deza and Elias B. Khalil. Machine learning for cutting planes in integer programming: A survey. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, IJCAI-2023, page 6592–6600. International Joint Conferences on Artificial Intelligence Organization, August 2023. URL <http://dx.doi.org/10.24963/IJCAI.2023/739>.
- [3] Gerald Gamrath, Benjamin Hiller, and Jakob Witzig. Reoptimization techniques for MIP solvers. In Evripidis Bampis, editor, *Experimental Algorithms - 14th International Symposium, SEA 2015, Paris, France, June 29 - July 1, 2015, Proceedings*, volume 9125 of *Lecture Notes in Computer Science*, pages 181–192. Springer, 2015. doi: 10.1007/978-3-319-20086-6_14. URL https://doi.org/10.1007/978-3-319-20086-6_14.

References II

- [4] Maxime Gasse, Didier Chételat, Nicola Ferroni, Laurent Charlin, and Andrea Lodi. Exact combinatorial optimization with graph convolutional neural networks. In *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pages 15554–15566, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/d14c2267d848abeb81fd590f371d39bd-Abstract.html>.
- [5] M. Güzelsoy. *Dual Methods in Mixed Integer Linear Programming*. PhD, Lehigh University, 2009. URL <http://coral.ie.lehigh.edu/~ted/files/papers/MenalGuzelsoyDissertation09.pdf>.
- [6] Aleksandr M. Kazachkov and Egon Balas. Monoidal strengthening of simple $\mathcal{V}$ -polyhedral disjunctive cuts, 3 2025. URL <https://doi.org/10.1007/s10107-024-02185-x>.
- [7] Krunal Kishor Patel. Progressively strengthening and tuning mip solvers for reoptimization, 2024. URL <https://arxiv.org/abs/2308.08986>.

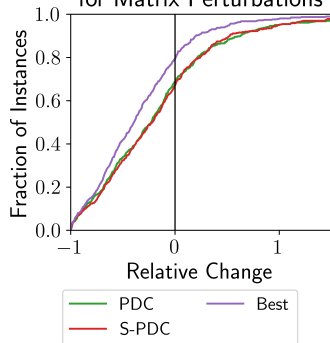
Paper 2: S/PDIs Strengthen Default Root Node Cut Generation

Degree	Terms	Element	VPC	S-PDI	PDI	Default
0.5	4	matrix	73.26	73.25	73.62	73.34
0.5	4	objective	61.21	–	61.20	60.07
0.5	4	rhs	51.76	48.47	49.65	50.20
0.5	64	matrix	73.58	73.59	73.40	73.42
0.5	64	objective	62.05	–	61.69	60.51
0.5	64	rhs	53.47	53.02	52.69	49.72
2.0	4	matrix	80.24	80.39	80.41	80.40
2.0	4	objective	61.84	–	62.34	61.06
2.0	4	rhs	52.80	52.45	52.89	53.77
2.0	64	matrix	80.61	80.24	79.39	80.12
2.0	64	objective	62.68	–	62.39	61.14
2.0	64	rhs	56.35	53.47	53.17	53.91

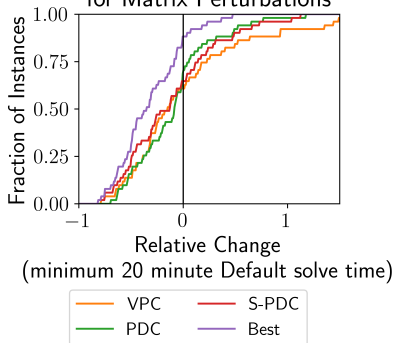
Takeaway: After default root cut generation, the clean strength ordering can blur because the solver's native cuts interact with the disjunctive generators and can wash out nominal differences in root-gap closure. For a defense, the key point is that this is not a dealbreaker: these cuts may still pay off deeper in the tree, where their pruning effect accumulates.

Paper 2: Branch and Cut Termination Time for Matrix Perturbations

Solve Time Change Relative to VPC
for Matrix Perturbations



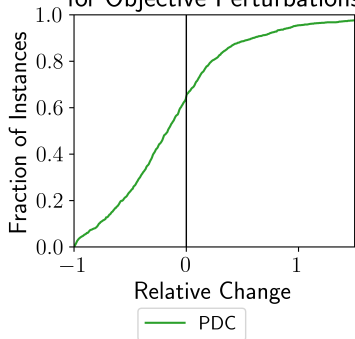
Solve Time Change Relative to Default
for Matrix Perturbations



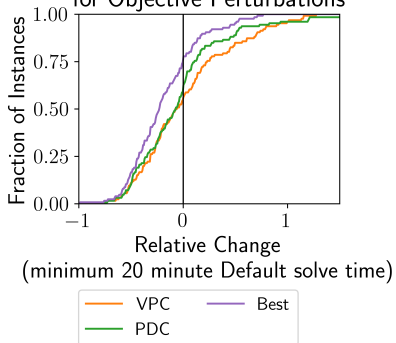
PDIs often outperform VPCs (left), and the best disjunctive cut generator almost always beats default Gurobi (right).

Paper 2: Branch and Cut Termination Time for Objective Perturbations

Solve Time Change Relative to VPC
for Objective Perturbations



Solve Time Change Relative to Default
for Objective Perturbations



PDIs often outperform VPCs (left) and default Gurobi (right).